

Enhancing recruitment strategies with a hybrid AutoML approach to employee promotion prediction

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ABSTRACT

This research proposes a hybrid Automated Machine Learning (AutoML) architecture for predicting employee promotions within Human Resource (HR) analytics to address the ongoing issue of class imbalance. A series of traditional machine-learning models and AutoML pipelines were evaluated comparatively alongside multiple resampling methods including SMOTE, SMOTE + Tomek Links, and SMOTE + ENN. In the proposed framework, the TPOT AutoML optimizer was integrated with interpretable learners, K-Nearest Neighbors and Extra Trees Classifier, through soft voting, which balances automation, accuracy, and interpretation. Results demonstrate that strong preprocessing methods and hybrid optimization methods improve predictive performance, achieving accuracy of up to 0.97 and F1 scores of 0.94 under SMOTE + ENN. Although ensemble models, such as Random Forest and LightGBM, performed equally well, the hybrid AutoML approach yielded better minority-class recall across the dataset. In addition to aggregate metrics, the study also illustrates potential HRIS-oriented deployment scenarios for explainable predictive analytics under controlled experimental settings. Despite the unavailability of actual HR data based on confidentiality, validated open data and cross-validation supported reproducibility and ethical considerations. This study presents an empirical evaluation of hybrid AutoML pipelines for promotion prediction under class imbalance, emphasizing integration rather than algorithmic novelty. Results are reported on a publicly available HR dataset and should be interpreted as illustrative of pipeline behavior under controlled conditions. The contribution lies in demonstrating how AutoML and interpretable ensemble models can be combined to improve robustness and minority-class recall, providing a reproducible baseline for future HR analytics research.

1. Introduction

Employee promotion is one of the critical aspects of effective talent management and helps develop the workforce, drive organizational success, and foster individual development. Effective promotion prediction enables HR departments to support timely and consistent decision-making; however, achieving reliable performance remains challenging in practice, particularly because standard machine learning models often require substantial domain expertise in feature engineering, algorithm selection, and hyperparameter tuning, which is frequently unavailable in HR departments.

Automated Machine Learning (AutoML) represents a paradigm shift in that it engages users through automation in the development and optimization of models to enable organizations to deploy these high-

performance predictive systems with limited or no technical involvement. A key challenge facing accurate promotion prediction relates to class imbalance—the different counts of promoted employees relative to non-promoted employee groups (Al-Alawi & Albuainain, 2024; Kediya et al., 2023). Other industries facing similar challenges have demonstrated the usefulness of hybrid resampling strategies and rigorous outlier removal processes—for example, in areas of fraud detection (Ounacer et al., 2020), medical diagnostics (Ardchir et al., 2023), and agriculture (Imade et al., 2024). Studies have shown great success using combinations including SMOTE + ENN, as a data-cleaning approach coupled with resampling, to improve classification performance for imbalanced datasets (Batista et al., 2004). Despite these advances, most existing approaches address imbalance or automation independently, with limited attention to their combined operational use in HR decision

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systems.

AutoML has been adopted across domains to automate pipeline construction and hyperparameter tuning, often achieving competitive performance with reduced manual effort (Castelli et al., 2022). However, AutoML pipelines are frequently treated as standalone solutions, with limited integration of interpretable learners or deployment-oriented HR decision-support design.

This study empirically evaluates an integration pipeline that combines TPOT AutoML, SMOTE-based balancing, and interpretable ensemble learners for promotion prediction. In the proposed framework TPOT-based AutoML is compared with standard algorithms, KNN and ExtraTreesClassifier, while exploring various data balancing methods including SMOTE, SMOTE+ENN, undersampling. The goal is to combine AutoML-driven pipeline optimization with interpretable learners in order to study performance trade-offs under different balancing strategies, rather than to introduce new learning algorithms.

This study situates predictive analytics within Human Resource Information Systems (HRIS) to illustrate how hybrid modeling pipelines can support promotion-related decision processes. The resultant insights can be translated into evidence-based recommendations for promotion, workforce planning, and talent management—ultimately converting algorithmic outcomes into actionable insights at the organizational level. The framework is intended to illustrate how AutoML-supported pipelines may be incorporated into HRIS environments to support transparent and reproducible promotion-related analytics.

After reviewing the literature and identifying research gaps within the scope of a publicly available HR dataset, the paper outlines the methodology, which includes data pre-processing, balancing approaches, as well as the hybrid framework implementation. Following the methodology, the experimental findings are presented, and afterwards, the authors provide a discussion of important findings, implications for HR analytics, and recommendations for future research.

2. Related works

Numerous studies have qualitatively examined the prediction of employee promotions using diverse machine learning methods aimed at maximizing prediction performance and ability to interpret the

prediction results (Thongkong, 2025; Abbour et al., 2025). Existing studies on employee promotion prediction can be broadly organized into three thematic streams: (i) imbalance-aware modeling that uses a resampling approach to help improve the ability to identify minority classes; (ii) hybrid/ensemble learning in which several classifiers are used in order to boost predictive accuracy; and (iii) AutoML (automated machine learning), which uses automated methods for designing and tuning the pipelines. Each of these three themes has been independently evaluated with respect to accuracy and robustness; however, there is still relatively little integration among automated model discovery (autoML), interpretable hybridization, and the resulting decision support system for HR departments in the area of employee promotion prediction. In Table 1, we summarize and provide a comparative analysis of the earlier articles methods, predictors and results discussed previously.

Early articles considered traditional algorithms and resampling approaches. For example, while using eight machine learning models, the highest accuracy was achieved with Extreme Gradient Boosting combined with a hybrid SMOTE + ENN resampling process, and also reported the factors region and the previous year’s rating as the most prominent predictors (Shafie et al., 2023). Although not a predictive modeling study, logistic regression and XGBoost were considered through the explainability tool LIME for enhanced explainability (Bhattacharya et al., 2023; Matin, 2025). Agrawal et al. (Agrawal et al., 2024) discussed an Agile-based framework for HR analytics, that was later experimentally validated on employee data (Alqahtani & Almaleh, 2022), but again highlighted gradient boosting as the most performant mode, with the previous rating as strong linearly dependent factor associated potential for many promotions. Finally, employing random forest models on a dataset with Chinese enterprises documented the years worked as a leading feature (Long et al., 2018).

Recent research has begun focusing on deep and ensemble learning techniques. A CNN–Random Forest hybrid produced 99 % as the highest accuracy, and was superior to a singular model (Ardehkhani et al., 2023). CatBoost achieved similarly good outcomes, allowing the processing of categorical variables and demonstrated interpretability (Rallapalli & Kumar, 2024). The Modified Additive Regression Classifier found an accuracy of 95.3 %, beating XGBoost, Support Vector Machines

Table 1
Comparative analysis of studies.

Study	Methods Used	Key Features/Techniques	Results	Insights
Shafie et al., 2023 (Shafie et al., 2023)	Eight ML algorithms, SMOTE+ENN	Key features: Region, previous year’s rating	High accuracy with Extreme Gradient Boost	SMOTE+ENN improves class balance; region and rating are critical predictors.
Bhattacharya et al., 2023 (Bhattacharya et al., 2023)	Logistic Regression, XGBoost	Explainable AI tools (LIME)	Interpretable predictions	LIME provides transparency in decision-making.
Agrawal et al., 2024 (Agrawal et al., 2024)	Agile methodology framework	Validated on HR datasets	High performance	Agile approaches enhance model development for HR analytics.
Alqahtani and Almaleh, 2022 (Alqahtani & Almaleh, 2022)	Gradient Boosting	Key feature: Prior ratings	Effective in promotion prediction	Feature importance improves interpretability.
Long et al., 2018 (Long et al., 2018)	Random Forest	Post-related features (e.g., working years)	Best performance on Chinese enterprise data	Post-related features highly influence promotions.
Ardehkhani et al., 2023 (Ardehkhani et al., 2023)	CNN-Random Forest hybrid model	Combination of CNN and Random Forest	99 % accuracy, outperforming standalone models	Hybrid models improve predictive performance.
Rallapalli and Kumar, 2024 (Rallapalli & Kumar, 2024)	CatBoost	Categorical feature handling	Superior accuracy and interpretability	Effective for datasets with categorical variables.
Agrawal and Goyal, n.d. (Agrawal & Goyal, 2023)	Modified Additive Regression Classifier	Comparison with XGBoost, SVM, Random Forest	95.30 % accuracy	Outperformed traditional ML methods.
Gong, 2023 (Gong, 2023)	Hybrid ANN-SVM model	ANN-SVM integration	Improved accuracy by 0.15 % to 7.26 %	Hybridization enhances prediction accuracy.
Al-Alawi and Albuaainain, 2024 (Al-Alawi & Albuaainain, 2024)	Random Forest, SMOTE	SMOTE and normalization techniques	99 % accuracy	Balancing techniques crucial for imbalanced data.
Tusell-Rey et al., 2024 (Tusell-Rey et al., 2024)	Evaluative Customized Naïve Associative Classifier	Evaluative Artificial Intelligence approach	Fairness and transparency in HR decisions	AI aligns predictions with fairness objectives.
Pan, 2020 (Pan, 2020)	AutoML	Simplification of recommendation algorithms	Improved generalization performance	AutoML reduces complexity in HR analytics.
Vijayan, n.d. (Vijayan, 2023)	Cloud-based tools (AWS Glue, SageMaker)	Automated feature engineering	Enhanced accuracy and scalability	Cloud platforms optimize HR predictive frameworks.

(SVM), and Random Forest and stressing the importance of model hybridization (Agrawal & Goyal, 2023). As with an ANN model (Gong, 2023), SVM performance improved by 0.15 % –7.26 % over performance by itself. Furthermore, data imbalance was addressed through SMOTE, and normalization and Random Forest accuracy was nearly perfect with 99 %, demonstrating balanced sampling is important in workforce human resource analytics (Al-Alawi & Albuainain, 2024). In addition to accuracy, some research has included principles of fairness and transparency. The Evaluative Customized Naïve Associative Classifier combined the Evaluative Artificial Intelligence framework to offer ethical, fair, and explainable promotion decisions (Tusell-Rey et al., 2024; Liu & Liu, 2024).

Recently, Machine Learning has transitioned into Automated Machine Learning (AutoML). AutoML facilitates algorithm selection and generalization for HR recommender systems (Pan, 2020; Bonthu, 2024). Simultaneously, cloud based predictive systems based on the AMAZON solution—AWS Glue and SageMaker—automated feature engineering and stable workforce predictive analytics was utilized (Vijayan, 2023).

Collectively, prior studies highlight three consistent trends: (1) advanced data-balancing strategies are essential for mitigating class imbalance; (2) hybrid and ensemble models generally outperform single classifiers; and (3) automation and interpretability are increasingly emphasized in HR analytics. However, most existing works address these dimensions separately—either focusing on resampling, hybrid modeling, or AutoML optimization—without empirically examining their joint integration within a unified and deployable HR decision framework. In contrast, the present study evaluates a hybrid pipeline that explicitly combines TPOT-based AutoML, class-balancing strategies, and interpretable ensemble learners, with attention to operational integration within Human Resource Information Systems (HRIS). This positioning differentiates the current work as an empirical investigation of integration logic rather than a proposal of new learning algorithms.

3. Conceptual framework of the proposed model

Following the results from the earlier sections of this research, this section presents the conceptual design and analytical pipeline. The proposed hybrid framework is presented in Fig. 1 and represents a hybrid framework that reflects the systematic pipeline observed in this study. The first step taken in the pipeline was the acquisition of data from Kaggle's "Employees Evaluation for Promotion" dataset. The

pipeline consisted of a series of important stages that are explained below.

First, exploratory data analysis (EDA) was conducted to visualize the distributions and relationships of variables. Missing values in the variables previous_year_rating and avg_training_score were imputed by allocating a value of 0 for new recruits to previous_year_rating and the mode for avg_training_score. Categorical encoding was then performed, the nominal variables were one-hot encoded and the ordered features were ordinal encoded. Numerical variables were then standardized using StandardScaler, and in the feature selection step, non-informative variables such as EmployeeCount and Over18 were removed from the data.

To address the class imbalance within the response variable, several resampling strategies were used including: SMOTE, SMOTE + Tomek Links, SMOTE + ENN, and random undersampling.

The next step was model training and tuning using AutoML (via TPOT), parameters were set to a population size of 100 and generation size of 10 to optimize for F1-score, and traditional machine learning algorithms including Logistic Regression, Random Forest, XGBoost, Gradient Boosting Classifier, and SVM. Finally, each models hyper-parameters were tuned using RandomizedSearchCV with 5-fold cross validation.

Finally, the model's performance was assessed using five metrics: accuracy, precision, recall and F1-score. In the following sections, we provide comparative evaluations between AutoML and traditional modeling methodologies under various types of data-balance methodologies.

4. Methodology

4.1. Data collection and description

This study employs the dataset "Employees Evaluation for Promotion," which is available online from Kaggle (Zaman). From a human resource management perspective, it encompasses employee characteristics and performance indicators relating to their promotion status in the course of one year. The dataset can serve as the foundation of predictive models to help HR departments identify employees that are most likely to be promoted and to assist in evidence-based decision making regarding promotions.

Table 2 summarizes the dataset, including its salient features, type of

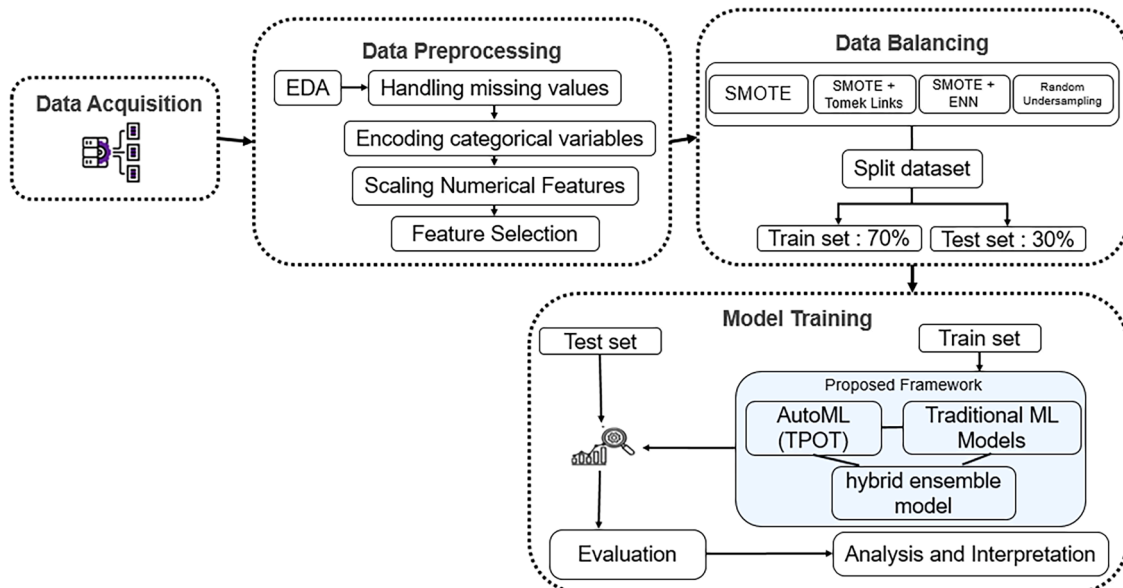


Fig. 1. Conceptual overview of the proposed framework.

Table 2

HR employee data overview.

Characteristic	Type	Example
Employee_id	Integer	65,438
Department	String	Sales & Marketing
Region	String	region_7
Education	String	Master's & above
Gender	String	f
Recruitment_channel	String	sourcing
No_of_trainings	Integer	1
Age	Integer	35
Previous_year_rating	Integer	5

variables, and example values. The dataset possesses 54,808 records and contains 13 features. The target variable, *is_promoted*, indicates whether the employee was promoted (1) or not (0). The primary features include:

- **Employee_id:** A unique identifier assigned to each employee.
- **Department:** The department to which the employee belongs.
- **Region:** The geographical region of the employee's assignment.
- **Education:** The education level of the employee.
- **Gender:** The gender of the employee.
- **Recruitment_channel:** The recruitment channel through which the employee was hired.
- **No_of_trainings:** The number of additional training programs the employee attended in the past year.
- **Age:** The age of the employee in years.
- **Previous_year_rating:** The performance rating of the employee from the preceding year.
- **Length_of_service:** The total years of service the employee has completed in the organization.

- **Awards_won:** A binary feature indicating whether the employee received any awards in the past year (1 for yes, 0 for no).
- **Avg_training_score:** The average score the employee achieved in recent training evaluations.

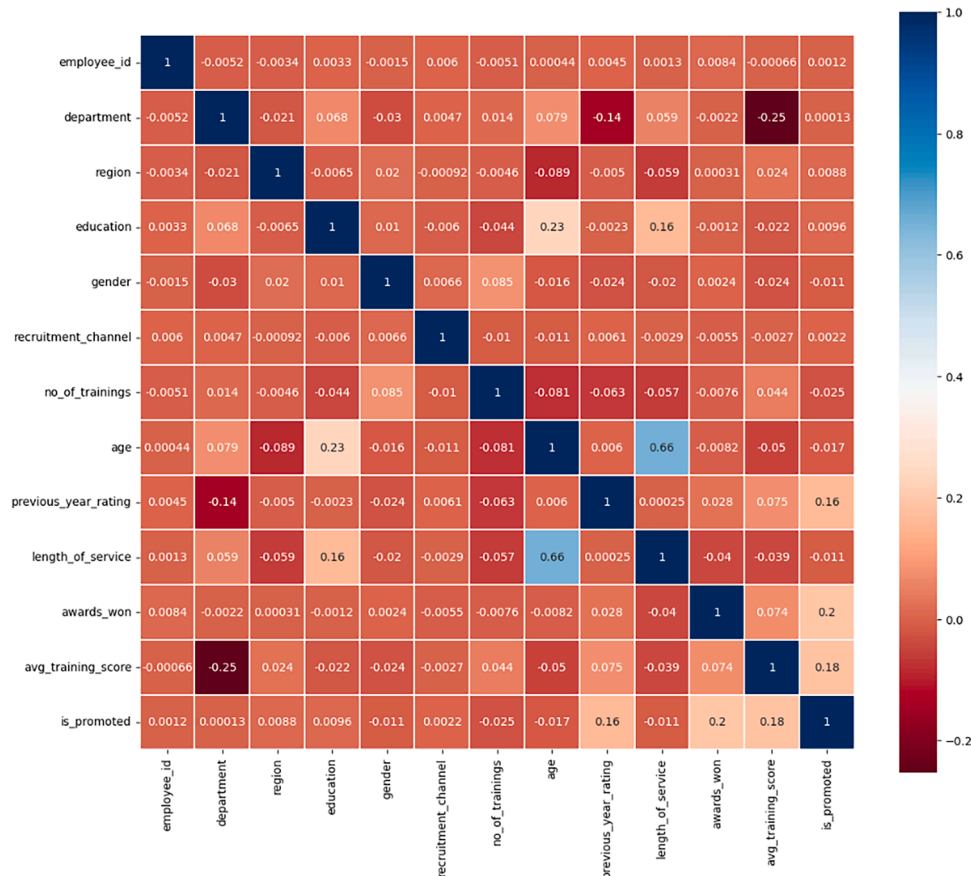
Every feature is crucial in evaluating employee potential and eligibility for promotion. The dataset contains both numerical and categorical variables, the data needed comprehensive preprocessing to ensure data quality and usability for machine-learning algorithms.

4.2. Data preprocessing

Data preprocessing was applied in a systematic manner to ensure the dataset integrity and suitability for modeling. The preprocessing pipeline followed standard practices summarized below. Exploratory Data Analysis (EDA) was performed to understand the dataset structure, to conduct descriptive analysis to understand trends in the data and to examine the features' relationship. The Pearson correlation matrix was visualized with a heatmap to determine the strength of the association between features. As displayed in Fig. 2, the correlation analysis indicated the following notable patterns:

Positive correlations were observed between *awards_won* and *avg_training_score* with *is_promoted*, suggesting that higher training performance and recognition are associated with greater promotion likelihood. Conversely, *no_of_trainings* exhibited minimal correlation with *is_promoted*, indicating limited influence on promotion decisions.

Handling Missing Values: Missing data in key features were treated as follows:

**Fig. 2.** Correlation matrix.

- Previous_year_rating: Missing values—mainly from employees with only one year of service—were imputed with 0, representing the absence of prior evaluation.
- Avg_training_score: Missing values were imputed using the mode to preserve data distribution.
- The imputation strategy combined statistical reasoning with domain logic to maintain representativeness and reproducibility.

Categorical Encoding and Transformation:

- One-Hot Encoding: Applied to nominal categorical features such as department, gender, and recruitment_channel to avoid ordinal bias.
- Ordinal Encoding: Used for ordered variables such as education and region to preserve inherent hierarchies.

Feature Scaling and Selection:

- Scaling: Numerical variables were standardized using the StandardScaler to ensure uniform feature magnitude, particularly for distance-based algorithms.
- Selection: Non-informative identifiers, such as employee id, were removed as they hold no predictive relevance.

Numerical features were standardized to ensure compatibility with distance-based classifiers.

4.3. Data balancing

The dataset exhibited a significant class imbalance, as illustrated in Fig. 3. The majority class (employees not promoted) constitutes 91.48 % of the dataset, with 50,140 instances, while the minority class (promoted employees) represents only 8.52 %, with 4668 instances.

Such a pronounced imbalance can lead to biased predictions, where the model tends to favor excessively the majority class, consequently undermining the accuracy and fairness of the predictive system. To address this challenge, the following resampling techniques were applied to balance the dataset:

- SMOTE (Synthetic Minority Over-sampling Technique): This method generated synthetic samples for the minority class by interpolating

existing instances, thereby increasing the representation of promoted employees without duplicating data.

- SMOTE + Tomek Links: This hybrid approach combined SMOTE with Tomek Links, which eliminated noisy and borderline samples that could confuse the classifier, resulting in improved class separation.
- SMOTE + ENN (Edited Nearest Neighbors): SMOTE was further enhanced with ENN, which refined the dataset by removing instances that were misclassified by their nearest neighbors, ensuring a cleaner training set.
- Random Undersampling: This technique reduced the number of majority class instances, balancing the dataset by ensuring fair representation of both classes, though at the cost of losing some data from the majority class.

These strategies for achieving balance became instrumental in reducing the negative impact of class imbalance and providing fairer and more accurate predictions of employee promotions. The selection of techniques considered how we could optimize the trade-off between maximizing the model performance while checking the robustness of training data. We selected these resampling strategies because they effectively combine oversampling and cleaning steps effectively for noisy, tabular HR datasets to create a balanced distribution while minimizing information loss and avoiding model bias to the majority class.

4.4. Model development

The proposed approach combines Automated Machine Learning (AutoML) and traditional algorithms in a hybrid framework meant to maximize the performance of promotion prediction. AutoML was conducted using TPOT (Tree-based Pipeline Optimization Tool), which uses genetic programming to automatically create and tune machine learning pipelines. TPOT considers a large number of combinations of pre-processing steps, feature selection methods, model algorithms, and hyperparameters to reach the best configurations. In this case TPOT was set with a population size of 100 and 10 generations, optimizing pipelines for the F1-score which values precision and recall despite class-imbalance.

At the same time, benchmark models were built using scikit-learn to

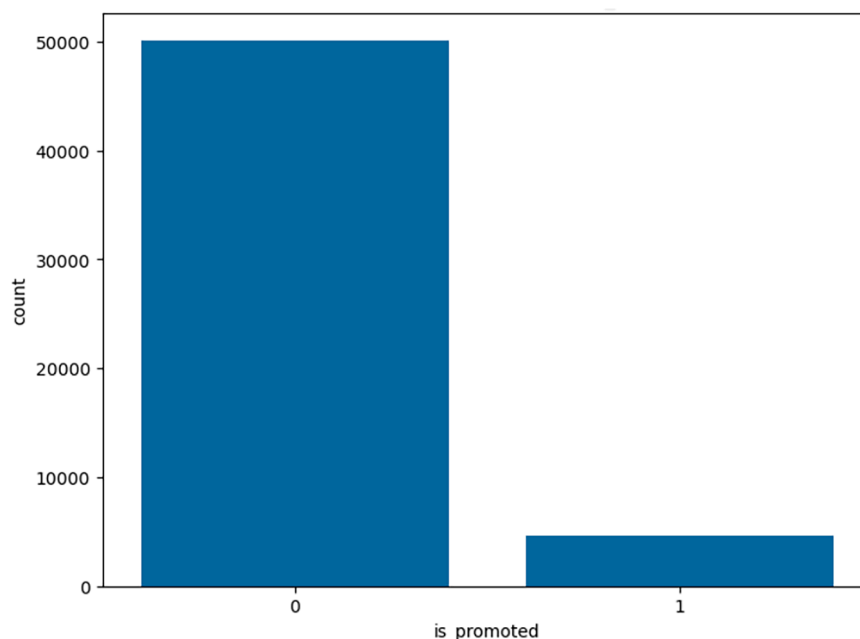


Fig. 3. Class imbalance in the promotion dataset.

enable systematic comparisons between the automated and manually tuned models. The traditional models were Logistic Regression, Random Forest, XGBoost, and Support Vector Machine (SVM). Like the AutoML pipelines, all models underwent hyperparameter tuning with RandomizedSearchCV, based on 5-fold cross validation for consistency and reproducibility over different balanced datasets. Results were compared across multiple resampling methods, SMOTE, SMOTE + Tomek Links, SMOTE + ENN, and Random Undersampling, to examine consistency and robustness.

The comprehensive hybrid framework is depicted in Fig. 4. The preliminary stages consist of data preparation followed by class balancing using SMOTE + Tomek Links was used to generate minority samples while reducing class overlap. This improves separability and reduces bias toward the majority class, a frequent issue in HR datasets with a limited number of employees who were promoted. TPOT AutoML is prompted to automatically discover an optimized baseline pipeline by choosing preprocessing processes, algorithm, and hyperparameters automatically, in an evolutionary search pattern. Although AutoML speeds up discovering pipelines, generalized optimization will not always account for task-specific irregularities. To mitigate the limitations of general optimization, two traditional models were introduced as complementary processes:

- K-Nearest Neighbors (KNN): captures local decision boundaries and neighborhood patterns.
- ExtraTreesClassifier: an ensemble tree-based learner offering robustness to noise and strong feature interaction modeling.

The final hybrid model combines the predictions made by three different base learners: the pipeline generated by TPOT, KNN, and ExtraTreesClassifier, using a soft-voting ensemble method. Each model outputs a probability, and at the end the probabilities are averaged to arrive at the final prediction. An equally weighted approach was taken here to ensure that all three learners are able to impact the final prediction equally, and a soft voting approach was selected over hard voting to retain information about the probabilities as compared to simply a class prediction. Equal weights were chosen to ensure that no dataset-specific biases would be imposed through manual adjustment of weights, while also creating consistent transparency across all base learners. Preliminary sensitivity tests using different weight assignments produced inconsistent performance gains; therefore, equal weights were retained as the default method for robustness and interpretability within the ensemble.

4.5. Hyperparameter optimization and cross-validation

Each of the traditional algorithms was optimized using RandomizedSearchCV with five-fold stratified cross-validation in order to ensure robust generalization on imbalanced classes. Ranges included, for example, n_estimators (50–300), and max_depth (3–15) for tree-based solutions and n_neighbors (3–15) for KNN. This was a compromise between computation time and maximum depth of exploration, while allowing for reproducible optimization. In the AutoML aspect, we ran

TPOT with a population size of 100 for 10 generations, optimizing the F1-score to encourage balanced predictive performance. The same random seeds and folds were matched across the experiments for consistency.

4.6. Evaluation metrics

Model performance was evaluated using multiple metrics to ensure a comprehensive assessment. The following metrics were used:

- Accuracy: Measures the proportion of correctly predicted instances out of the total instances.

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN} \quad (1)$$

- Precision: Evaluates the reliability of the model in identifying promotions by calculating the proportion of true positive predictions out of all positive predictions.

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

- Recall (Sensitivity): Measures the model's ability to correctly identify actual promotions by calculating the proportion of true positives out of all actual positives.

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

- F1-score: The harmonic mean of precision and recall, prioritizing a balance between them. It is particularly effective for datasets with class imbalance.

$$F1 - score = \frac{2}{\frac{1}{precision} + \frac{1}{recall}} \quad (4)$$

Among these metrics, accuracy was given the most emphasis when comparing the different models, as it is a global measure of total instances correctly classified. Accuracy metric used for a general evaluation of model performance was appropriate for the data set characteristics, including the distribution of promoters and non-promoters.

All metrics were calculated from the held-out test folds and averaged across five random seeds so that all reported results represent stable, reproducible performance, and not variability from a single run.

5. Experiment and results

5.1. Comprehensive evaluation under class imbalance

All model performances were averaged using five-fold stratified

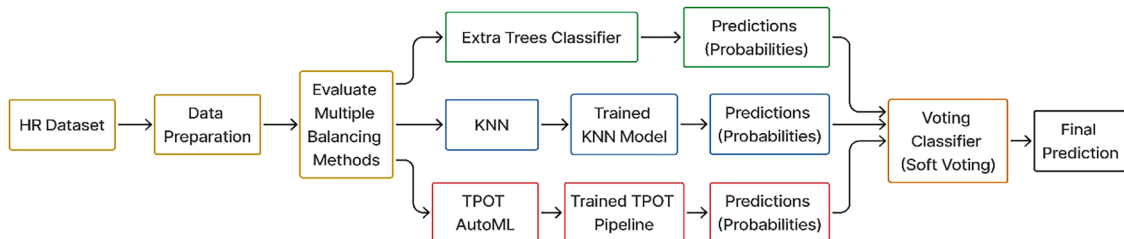


Fig. 4. The proposed Hybrid framework for promotion prediction.

cross-validation to ensure the robustness of results with class imbalance as well as to reduce overfitting. The evaluation results for four representative models, the classical baselines (Logistic Regression and Random Forest), the TPOT AutoML pipeline, and the proposed Hybrid Ensemble, are given in Table 3.

The Hybrid Ensemble's voting model synthesizes the TPOT pipeline's optimization of hyper-parameters with two other complementary learners: K-Nearest Neighbors (KNN) and Extra Trees Classifier. This hybrid structure achieved this fusion between the automated hyper-parameter optimization of AutoML while maintaining the interpretability and stability of classical algorithms. In addition to accuracy, the evaluation metrics included precision, recall, and F1-score for a complete evaluation of the predictive reliability of the data sets tested for class imbalance. The TPOT AutoML pipeline had the highest overall accuracy (0.94) compared to the Hybrid Ensemble model, which produced slightly less balanced precision and recall, but an F1-score of 0.94 and recall of 0.92. This indicates that the Hybrid Ensemble produced a more sensitive detection rate to the minority class of promoted employees; consistency with the TPOT AutoML pipeline overall surge in accuracy. Conversely, Random Forest showed higher precision than recall, indicating a more conservative prediction approach for adaptive elements.

From an applied perspective, the Hybrid Ensemble increases recall for the minority class while maintaining competitive overall accuracy. However, the practical value of these gains depends on organizational decision thresholds and the costs of false positives versus false negatives. Therefore, we report these results as empirical improvements under the evaluated dataset settings rather than as direct evidence of operational impact in HRIS deployments.

5.2. Comparative analysis

The experimental design evaluated models with multiple resampling methods: SMOTE, SMOTE + Tomek Links, SMOTE + Edited Nearest Neighbors (ENN), and Random Undersampling. The models were organized into methodological families which were Bagging, Boosting, Probabilistic, Others, Linear, Neural Networks, and Automated ML. Table 4 showed resulting accuracies and Fig. 5 showed the comparative performance. Models utilizing Bagging like Random Forest and Extra Trees reached the highest accuracies (≈ 0.95) when utilizing over-sampling (and both types of SMOTE). Likewise, Boosting models (XGBoost, LightGBM) had good performances, but had low levels of accuracies when utilizing undersampling methods. Both Linear and probabilistic models saw slight improvement when utilizing SMOTE however their overall performances were the weakest. The TPOT AutoML and hybrid framework generated consistently high results on all but two balancing methods and saw maximum performance (0.97) when utilizing SMOTE + ENN. These results indicate that the hybrid model remains robust across multiple resampling configurations within the evaluated dataset.

5.3. Analysis and interpretation

The application of balancing techniques greatly enhanced model accuracy across most algorithms. SMOTE and its variations regularly

Table 3
Model performance across key metrics.

Model	Accuracy	Precision	Recall	F1-Score	Training Time (s)
Logistic Regression	0.92	0.88	0.84	0.86	15
Random Forest	0.93	0.92	0.87	0.89	35
TPOT AutoML	0.94	0.93	0.90	0.91	120
Hybrid Ensemble	0.94	0.92	0.92	0.94	140

performed better than training the model unbalanced, confirming the advantages of generating synthetic minority examples, as well as noise reduction. On the other hand, the use of random undersampling caused a significant drop in accuracy due to lost information. Among the balancing approaches, bagging models (specifically, Random Forest and Extra Trees) were the most stable and high-performing across all approaches (i.e., approx. 0.95 accuracy). The boosting algorithms (LightGBM and XGBoost) displayed very slight variations in performance with some methods in which they remain competitive in results. Correspondingly, the probabilistic models (Naive Bayes) and linear classifiers struggled to adjust-adapt to samples produced synthetically with little improvement. The artificial neural network improved just slightly with SMOTE + ENN but continued to lag behind the tree-based ensembles. The TPOT AutoML pipelines were produced by the best models in terms of accuracy- consistently performing as well, or better than traditional bagging models at 0.95.

More specifically, the hybrid framework proposed produced the best overall accuracy (0.97 ± 0.01) and F1 (0.94 ± 0.02) across all SMOTE variants and clearly shows that the approach of combining an AutoML optimization procedure, and then manual refining of the second ensemble, provides benefit to both balance and robustness across resampling settings within the dataset. In the models proposed, cross-validation, regularization, and early-stopping controls supported that the results were consistent and limited overfitting. While the Kaggle dataset builds limitations on the ability to generalize to real-world application, the results from across folds provides promising results for adaptation to HRIS (Human Resource Information Systems).

While the proposed hybrid framework achieves higher accuracy and F1-score compared to baseline models, the observed improvements should be interpreted cautiously. Performance gains, although consistent across resampling strategies, range between 2 %–4 % for most metrics. These differences indicate incremental rather than transformative improvements. Such margins are common in mature classification tasks and reflect optimization of pipeline integration rather than radical algorithmic advances.

The Hybrid Ensemble provides the highest accuracy reported to date at 0.97 and F1 score at 0.94, which are substantial improvements over the strong baseline models of Random Forest and TPOT. However, when looking at the numbers in absolute terms, they are not that large. Therefore, the hybrid can be seen as providing a new level of stability across multiple resampling techniques, as well as higher minority class recall, rather than a marked improvement in total accuracy.

We did not conduct formal hypothesis testing between any models and, consequently, we advise interpreting differences descriptively through cross-validation means.

From a practical standpoint, while small improvements in promoted employee recall (HR) can make a significant difference in terms of morale and retention and talent growth in an HR environment, the extent of these improvements are dependent upon the context of the company and threshold for making decisions related to promotion.

5.4. Comparison of approaches

The table below compares Traditional Machine Learning, AutoML, and the proposed Hybrid Approach, with respect to several salient aspects such as automation, flexibility, ease of use, performance, imbalanced data handling, efficiency, interpretability, use case scenario, and scalability. The summary of those aspects is outlined in detail in Table 5. It is important to understand these compromises to select the most suitable approach for a given machine learning task.

Traditional ML provides flexibility and interpretability in results from algorithms, but significant expertise is needed. AutoML provides automation, efficiency, and lower barriers in terms of usability, but it may suffer from reduced flexibility and interpretability. The hybrid approach includes both the traditional and automated ML approaches achieved through the extended evaluation pipeline, allowing the data

Table 4
Comparative analysis of model accuracy under different data-balancing techniques.

category	Model	Unbalanced Data	SMOTE	SMOTE + Tomek	SMOTE + ENN	Undersampling
Bagging	Random Forest	0.93	0.95	0.95	0.95	0.70
	Extra Trees Classifier	0.93	0.95	0.95	0.95	0.69
Boosting	XGBoost	0.94	0.93	0.92	0.93	0.70
	AdaBoost	0.92	0.82	0.82	0.82	0.69
	LightGBM	0.94	0.93	0.93	0.92	0.72
	Naive Bayes	0.91	0.67	0.67	0.70	0.61
Probabilistic Models	Decision Tree	0.88	0.91	0.92	0.93	0.64
Others	K-Nearest Neighbors (KNN)	0.92	0.87	0.87	0.94	0.63
	SVM	0.91	0.69	0.69	0.72	0.63
	Logistic Regression	0.92	0.69	0.68	0.71	0.67
Linear Models	Artificial Neural Network (ANN)	0.93	0.80	0.80	0.83	0.68
Neural Networks	TPOT	0.94	0.95	0.95	0.95	0.72
Automated ML	TPOT	0.94	0.97	0.97	0.97	0.69

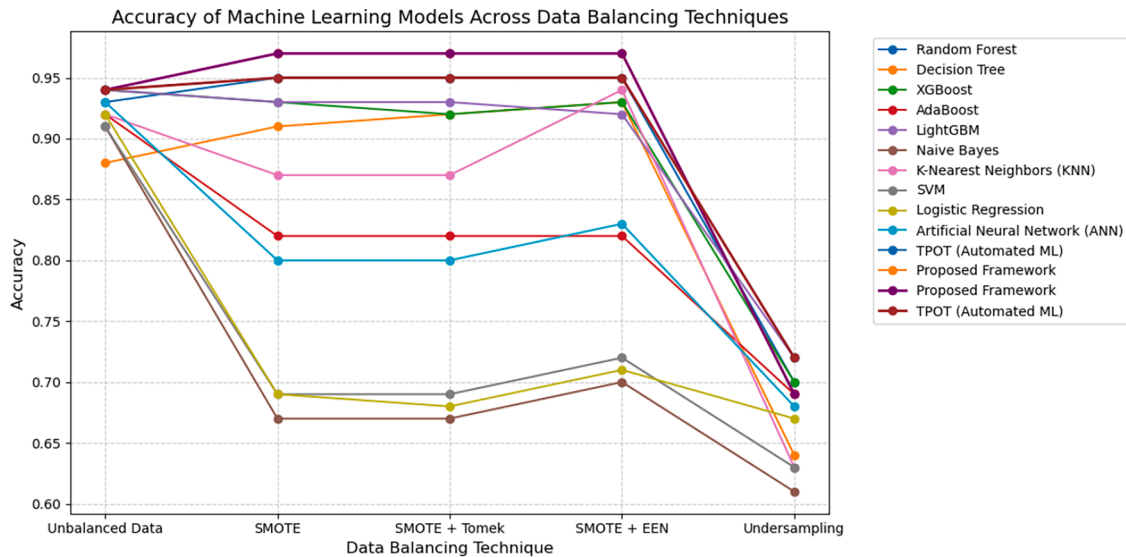


Fig. 5. Comparative analysis of model accuracies under different resampling techniques.

scientist to achieve high performance, interpretability, and adapting to changing stakeholder domains and conditions. The hybrid approach seems particularly suited for contexts of predicting HR promotions.

6. Discussion

Our experimental results indicate that through the use of AutoML-driven pipeline optimization operations, when used together with complementary interpretable learners, provides the best stability across resampling strategies and maximum recall in minority classes while providing only tentative increases in overall accuracy. This indicates that the major benefit of using this hybrid approach is its ability to be robust when faced with class imbalance, in addition to having operational transparency, rather than providing additional predictive capabilities. In the context of information management; these findings suggest that hybrid pipelines are best suited for integration as decision support components of an HRIS, such as ranking or recommendation modules, rather than as fully automated decision-making systems. As such, probabilistic outputs from the hybrid pipelines can enable configurable decision thresholds in line with an organization's risk preference (e.g., trade-offs between recall and precision). Furthermore, the integration of hybrid pipelines into HRIS should incorporate governance mechanisms to ensure auditing capability for model versioning and feature inputs, as well as regular ongoing monitoring for potential drift in feature input distributions and regular update policies for retraining and continued reliability over time.

The methodology outlined in this framework provided proof-of-concept for hybridizing Automated Machine Learning into an existing HR Information System. Results from these experiments should be interpreted as indicating trends and examples of what is possible when using hybridized machine learning systems but should not be viewed as conclusive findings. The research presented in this article was conducted using only a single publicly available humane resource dataset (that is, HRD) and does not provide representation of the full range of variables (for example, organization, culture, policies, etc.) that are likely to exist in an actual HRIS implementation. Therefore, the results indicate trends for a specific dataset and cannot be generalized to any other datasets without additional validation. Follow-up studies should replicate and verify the findings of this study across different HRD datasets and also assess any operational constraints associated with deploying AutoML products in real-world environments.

While this research has demonstrated practical solutions for promoting employees, building on the premise of a hybridised AutoML-integrated solution using human-interpretable ensemble learning models, this theoretical framework has primarily focused on exploring robust integration into operational systems. Future research could include developing a robust, actionable path for creating governance-accountability compliant and explainable decision-support systems.

7. Conclusion

This study illustrates the possibilities of data-driven methodologies

Table 5
Comparison of traditional machine learning, AutoML, and Hybrid approaches.

Aspect	Traditional ML	AutoML	Hybrid Approach
Automation	Requires manual selection of features, algorithms, and hyperparameters.	Fully automated, including feature selection, model selection, and hyperparameter tuning.	Combines automation (from AutoML) with manual fine-tuning for specific models.
Flexibility	High flexibility; users can customize every step but requires expertise.	Limited to the tools and methods supported by the AutoML framework.	Flexible; combines automated processes with user-selected additional models.
Ease of Use	Requires significant knowledge and effort for implementation.	User-friendly; minimal expertise required to set up and run.	Moderate; requires expertise for manual additions but simplifies initial steps.
Performance	Dependent on the user's expertise and ability to tune models.	Typically good for standard datasets and tasks but may miss nuanced improvements.	High performance; leverages AutoML's strengths and complements with specialized models.
Handling Imbalanced Data	Requires manual implementation of balancing techniques like SMOTE.	Often includes automated balancing or optimization for imbalanced datasets.	Integrates advanced balancing techniques like 'SMOTE + Tomek Links' alongside automation.
Efficiency	Time-consuming due to manual steps.	Fast and efficient; automates the entire pipeline.	Efficient; automation accelerates initial steps, and manual intervention optimizes further.
Interpretability	High; users have full control and understanding of each step.	Moderate; the automated process can feel like a black box.	Balanced; interpretable results from manual models enhance trust in automation.
Use Case Suitability	Ideal for small-scale projects or tasks needing domain expertise.	Best for standard machine learning problems with limited time/resources.	Suitable for complex problems requiring both automation and domain-specific customization.
Scalability	Scalable but labor-intensive for large datasets or multiple tasks.	Highly scalable; handles large datasets and tasks with minimal effort.	Scalable; hybridization ensures robustness and adaptability for large-scale tasks.

in tackling the critical issues presented in Human Resource (HR) analytics, specifically in regards to promotion prediction under class imbalance. Using a systematic investigation of traditional machine-learning models and AutoML, in conjunction with several resampling mechanisms, the study provides usable resources to exploit predictive frameworks, when applied within HR decision-making processes.

A proposed hybrid AutoML framework successfully brings together SMOTE-based data balancing with better modeling tuning. The best accuracies achieved, up to 0.97 across a range of SMOTE variants, indicate that appropriate preprocessing and careful tuning resulted in notable predictive gains. SMOTE + ENN produced the best trade-off

between recall and F1-score while maintaining reasonable accuracy in balancing strategy. While classical ensemble approaches, such as Random Forest and LightGBM, continue to perform well, the AutoML derived TPOT pipeline and suggested hybrid approach, consistently exhibited competitive performance across accuracy.

While predictive performance is an important consideration, so is interpretability and ethical governance. The framework may support transparent, traceable, and data-driven promotion decisions by ensuring the embedding of AutoML utilization in Human Resource Information Systems (HRIS). The implementation of AutoML within the HRIS creates an opportunity for predictive analytics to become a part of an organization's knowledge system which leads to continuous learning and ethical, and responsible, integration of AI.

While we utilized publicly accessible Kaggle data due to restricted access to enterprise HR records, we executed cross-validation and multi-metric assessments of our methods to warrant methodological reliability and reproducibility. Forthcoming explorations will broaden our validation of the framework to additional HR applications (e.g., employee retention, performance appraisal, and talent acquisition) and incorporate explainable AI methods as SHAP, LIME and fairness dashboards in conviction with available HRIS systems. These extensions will enhance transparency, scalability, and ethical and social accountability.

In summary, this work offers an incremental contribution by empirically evaluating a hybrid AutoML-ensemble pipeline for promotion prediction under class imbalance. In this study, we have shown that the results are consistent across multiple resampling techniques and have demonstrated some minor improvements in terms of recall for the minority class. Although this is based on a single public dataset, the work provides a usable baseline for reproducibility as well as some useful insights into the combination of AutoML and interpretive models in human resource analytics. Future efforts will need to evaluate this work within an organizational data set and apply official fairness measurements for a more thorough assessment of how this would apply in practice.

CRediT authorship contribution statement

Fatima Zahra Abbour: Writing – original draft, Visualization, Software, Resources, Methodology, Investigation, Formal analysis, Data curation. **El Mehdi Lghaouch:** Writing – review & editing, Resources, Investigation, Formal analysis, Conceptualization. **Khadija Mahjoubi:** Visualization, Resources, Data curation, Conceptualization. **Soufiane Ardchir:** Validation, Supervision, Methodology, Conceptualization. **Soumaya Ounacer:** Validation, Supervision, Methodology, Conceptualization. **Mohamed Azzouazi:** Validation, Supervision, Project administration, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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